Let $\mathcal{H}_K = \begin{pmatrix} \Delta & v(\pi_x - i\pi_y) \\ v(\pi_x + i\pi_y) & -\Delta \end{pmatrix}$ where $\Delta$ represents staggered potential which induces asymmetry between A and B sublattices.

Next $\begin{cases} v(\hat{\pi}_x - i\hat{\pi}_y) = \hbar\omega_c \hat{a} \\ v(\hat{\pi}_x + i\hat{\pi}_y) = \hbar\omega_c \hat{a}^+ \end{cases}$

$$\begin{pmatrix} \Delta & \hbar\omega_c \hat{a} \\ \hbar\omega_c \hat{a}^+ & -\Delta \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \varepsilon \begin{pmatrix} u \\ v \end{pmatrix} \Rightarrow \begin{cases} \hbar\omega_c \hat{a} v = (\varepsilon - \Delta) u \\ \hbar\omega_c \hat{a}^+ u = (\varepsilon + \Delta) v \end{cases}$$

$$\Rightarrow \hbar\omega_c \hat{a}^+ \frac{\hbar\omega_c}{\varepsilon - \Delta} \hat{a} v = (\varepsilon + \Delta) v$$

$$(\hbar\omega_c)^2 \underbrace{\hat{a}^+ \hat{a}}_{\hat{N}} v = (\varepsilon^2 - \Delta^2) v$$

Hence $\begin{cases} v = v_n = |n\rangle \text{ eigenstate of Harmonic oscillator} \\ \hat{N} v_n = n v_n \end{cases}$

$$\Rightarrow \varepsilon_n^2 = (\hbar\omega_c)^2 n + \Delta^2$$

$$\varepsilon_n = \pm\sqrt{\hbar^2\omega_c^2 n + \Delta^2}$$

Have to be careful about $n=0$ state:

$$v_{n=0} = |n=0\rangle = |0\rangle$$

$$u_{n=0} = \frac{\hbar\omega_c}{\varepsilon - \Delta} \hat{a} v_{n=0} = 0 \Rightarrow \hat{a}^+ u_{n=0} = 0 = (\varepsilon + \Delta) v_{n=0}$$

Thus, at point K: $\varepsilon^{K}_{n\neq 0,\lambda} = \lambda\sqrt{\hbar^2\omega_c^2 n + \Delta^2}$

where $\lambda = +1$ for $n = +1, +2, +3.$

$\lambda = -1$ $\quad n = -1, -2, -3 \ldots$

e.g. $\lambda = \mathrm{sgn}(n)$.

For $\underline{n=0}$ $\quad \varepsilon^{K}_{n=0} = -\Delta$ (one state)

Now look on K' point:

$$H_{K'} = \begin{pmatrix} \Delta & -v\pi_+ \\ -v\pi_- & -\Delta \end{pmatrix} = \begin{pmatrix} \Delta & -\hbar\omega_c \hat{a}^+ \\ -\hbar\omega_c \hat{a} & -\Delta \end{pmatrix}$$

$$\begin{pmatrix} \Delta & -\hbar\omega_c \hat{a}^+ \\ -\hbar\omega_c \hat{a} & -\Delta \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = \varepsilon \begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{cases} -\hbar\omega_c \hat{a}^+ v = (\varepsilon - \Delta) u \\ -\hbar\omega_c \hat{a}\, u = (\varepsilon + \Delta) v \end{cases}$$

$$v = -\frac{\hbar\omega_c}{\varepsilon+\Delta}\hat{a}\, u \Rightarrow -\hbar\omega_c \hat{a}^+\left(-\frac{\hbar\omega_c}{\varepsilon+\Delta}\hat{a}\right) u = (\varepsilon - \Delta) u$$

$$(\hbar\omega_c)^2 \hat{a}^+\hat{a}\, u = (\varepsilon^2 - \Delta^2) u$$

$$\Rightarrow u = u_n = |n\rangle$$

$$\varepsilon_n^2 = \Delta^2 + (\hbar\omega_c)^2 n$$

$$\varepsilon_n = \pm\sqrt{(\hbar\omega_c)^2 n + \Delta^2}$$

Again, look at $\underline{n=0}$:

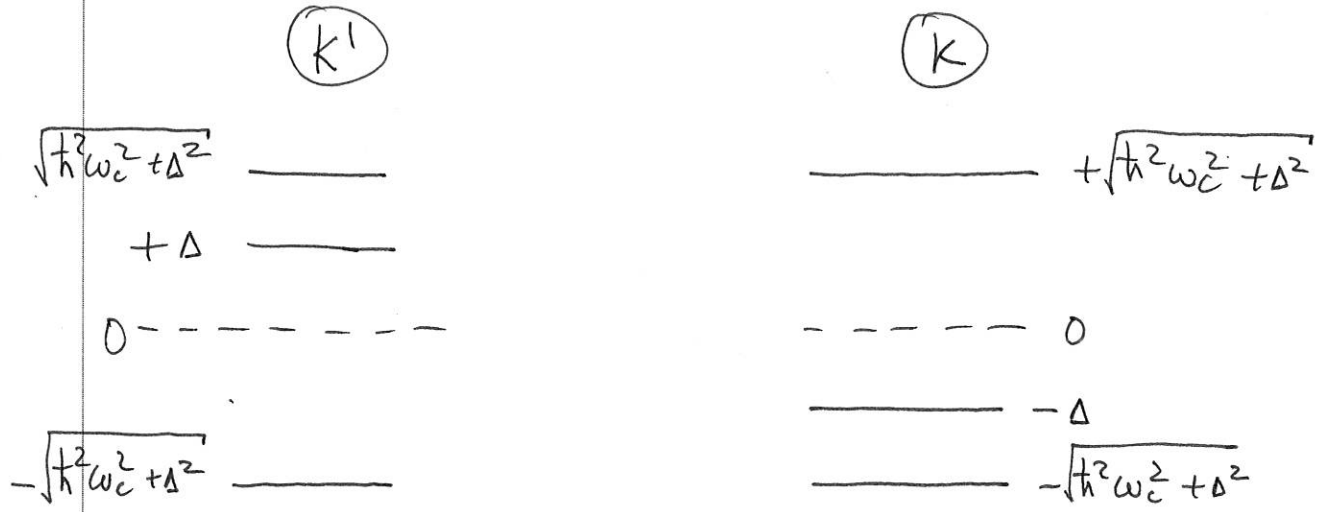
$$(\varepsilon + \Delta) v_{n=0} = -\hbar\omega_c \hat{a}\, u_{n=0} = 0$$

$$\Rightarrow \hat{a}^+ v_{n=0} = 0 = (\varepsilon - \Delta) u_{n=0} \Rightarrow \varepsilon_{n=0} = \Delta.$$

Thus, $\varepsilon^{K'}_{n\neq 0,\lambda} = \lambda\sqrt{\hbar^2\omega_c^2 n + \Delta^2}$ , $\lambda = \mathrm{sgn}(n)$

(Same as at K) $n = \pm 1, \pm 2, \pm 3 \ldots$

but $\varepsilon^{K'}_{n=0} = +\Delta$. (one state)




Observe that for $n \neq 0$ each of the states $\mathrm{sgn}(n)\sqrt{\hbar^2\omega_c^2 n + \Delta^2}$ occurs twice, while $+\Delta$ and $-\Delta$ ones appear only once.

Following Koshino and Ando (PRB 81 195431 (2010))

$$\varepsilon_\lambda(x_n) = \lambda\sqrt{x_n+\Delta^2} \quad , \; x_n = (\hbar\omega_c)^2 n \quad , \; \lambda = \pm 1$$

Thermodynamic potential

$$\Omega = -k_B T \underbrace{\frac{g_v g_s}{2\pi \ell_B^2}}_{\substack{\text{degeneracy of} \\ n\text{-th level } (n\neq 0)}} \sum_{\lambda, n} \ln\left(1+e^{\beta(\mu-\varepsilon_{\lambda,n})}\right)\underbrace{\left(1-\frac{1}{2}\delta_{n,0}\right)}_{\substack{\text{this takes} \\ \text{care of} \\ \text{factor (2)} \\ \text{smaller} \\ \text{degeneracy for} \\ \varepsilon_{n=0}=\pm\Delta \text{ levels.}}}$$

Euler-Maclaurin formula
(pp. 67' - 67'''')

$$\frac{1}{2}F(0) + \sum_{n=1}^{\infty} F(n) = \int_0^\infty dn\, F(n) - \frac{1}{12}F'(0) + \dots$$

$$\Omega = -k_B T \frac{g_v g_s}{2\pi \ell_B} \sum_{\lambda=\pm 1} \left\{ \frac{1}{2}\ln\left(1+e^{\beta(\mu-\varepsilon_{\lambda,n=0})}\right) + \sum_{n=1}^{\infty} \ln\left(1+e^{\beta(\mu-\varepsilon_{\lambda,n})}\right)\right\}$$

hence $F(n) = \ln\left[1+e^{\beta(\mu-\lambda\sqrt{\hbar^2\omega_c^2 n+\Delta^2})}\right]$

$$\Omega = -k_B T \frac{g_v g_s}{2\pi \ell_B^2} \sum_{\lambda=\pm 1} \left\{ \int_0^\infty dn \ln\left(1+e^{\beta(\mu-\lambda\sqrt{\hbar^2\omega_c^2 n+\Delta^2})}\right) - \frac{1}{12}\frac{\partial}{\partial n}\ln\left(1+e^{\beta(\mu-\lambda\sqrt{\hbar^2\omega_c^2 n+\Delta^2})}\right)\Bigg|_{n=0}\right\}$$

$$\int_0^\infty dn \, \ln\left(1 + e^{\beta\mu} e^{-\beta\lambda\sqrt{\hbar^2\omega_c^2 n + \Delta^2}}\right) =$$

$$= \int_0^\infty \frac{dx}{(\hbar\omega_c)^2} \ln\left(1 + e^{\beta\mu} e^{-\beta\lambda\sqrt{x+\Delta^2}}\right)$$

$$\hbar\omega_c = \sqrt{\frac{2e\hbar B v^2}{c}} \Rightarrow \frac{1}{(\hbar\omega_c)^2} = \frac{c}{2e\hbar B v^2};$$

Also, $\frac{\partial}{\partial n} \ln\left(1 + e^{\beta\mu} e^{-\beta\lambda\sqrt{\hbar^2\omega_c^2 n + \Delta^2}}\right)\Big|_{n=0} =$

$$= \frac{e^{\beta\mu} e^{-\beta\lambda\sqrt{\hbar^2\omega_c^2 n + \Delta^2}}}{1 + e^{\beta\mu} e^{-\beta\lambda\sqrt{\hbar^2\omega_c^2 n + \Delta^2}}} \left(\frac{-\beta\lambda \cdot \hbar^2\omega_c^2}{2\sqrt{\hbar^2\omega_c^2 n + \Delta^2}}\right)\Bigg|_{n=0} =$$

$$= -\frac{e^{\beta\mu} e^{-\beta\lambda\Delta}}{1 + e^{\beta\mu} e^{-\beta\lambda\Delta}} \frac{\beta\lambda}{2\Delta} \hbar^2\omega_c^2 .$$

$$\Rightarrow \Omega = -k_B T \frac{g_v g_s}{2\pi l_B^2} \sum_{\lambda=\pm} \left\{ \frac{c}{2e\hbar B} \int_0^\infty dx \, \ln\left(1 + e^{\beta\mu} e^{-\beta\lambda\sqrt{x+\Delta^2}}\right) \right.$$

$$\left. + \frac{1}{12} \frac{\lambda \beta \hbar^2\omega_c^2}{2\Delta} \frac{e^{\beta\mu} e^{-\beta\lambda\Delta}}{1 + e^{\beta\mu} e^{-\beta\lambda\Delta}} \right\}$$

Coefficient of integral: $\frac{g_v g_s}{2\pi l_B^2} \frac{c}{2e\hbar B v^2} = \frac{g_v g_s}{2\pi\left(\frac{\hbar c}{eB}\right)} \cdot \frac{c}{2e\hbar B} = \frac{g_v g_s}{4\pi\hbar^2 v^2}$

is B-independent

$\Rightarrow$ (just as for usual ll. gas), the first term does not depend on B.

2nd term: $k_B T \dfrac{g_v g_s}{2\pi l_B^2} \beta \dfrac{2e\hbar B v^2}{c} = \dfrac{g_v g_s}{\pi} \dfrac{eB}{\hbar c} \dfrac{e\hbar B v^2}{c} =$

$$= \frac{g_v g_s e^2 v^2}{\pi c^2} B^2$$

Thus $$\Omega = \Omega(B=0) - \sum_{\lambda=\pm 1} \frac{g_v g_s e^2 v^2 B^2}{12\pi c^2} \frac{\lambda}{2\Delta} \frac{e^{\beta\mu} e^{-\beta\lambda\Delta}}{1+e^{\beta\mu}e^{-\beta\lambda\Delta}}$$

$$= \Omega(B=0) - B^2 \frac{g_v g_s e^2 v^2}{12\pi c^2 \Delta} \left( \frac{e^{\beta\mu} e^{-\beta\Delta}}{1+e^{\beta\mu}e^{-\beta\Delta}} - \frac{e^{\beta\mu} e^{\beta\Delta}}{1+e^{\beta\mu}e^{\beta\Delta}} \right)$$

$$= \Omega(B=0) - B^2 \frac{g_v g_s e^2 v^2}{12\pi c^2 \Delta} e^{\beta\mu} \frac{e^{-\beta\Delta}(1+e^{\beta\mu}e^{\beta\Delta}) - e^{\beta\Delta}(1+e^{\beta\mu}e^{-\beta\Delta})}{1+e^{2\beta\mu}+2e^{\beta\mu}\mathrm{ch}(\beta\Delta)}$$

$$= \Omega(B=0) + \frac{g_v g_s e^2 v^2}{12\pi c^2 \Delta} (B^2) \frac{2\sinh(\beta\Delta)}{2\cosh(\beta\mu)+2\cosh(\beta\Delta)}$$

Thus magnetization $M = -\left(\dfrac{\partial \Omega}{\partial B}\right)_\mu$

$$M = -\frac{g_v g_s e^2 v^2 B}{6\pi c^2 \Delta} \frac{\sinh(\beta\Delta)}{\cosh(\beta\Delta)+\cosh(\beta\mu)}$$

Susceptibility

$$\chi = \frac{\partial M}{\partial B} = -\frac{g_v g_s e^2 v^2}{6\pi c^2} \cdot \frac{\sinh(\beta\Delta)}{\Delta\left[\cosh(\beta\Delta)+\cosh(\beta\mu)\right]}$$

$\left(\beta \equiv \dfrac{1}{k_B T}\right)$

diamagnetic susceptibility!

Now we can investigate various limits:

① $\Delta \to 0$ (so as to obtain the case of ideal graphene)

$$\lim_{\Delta \to 0} \frac{\sinh(\beta\Delta)}{\Delta} = \beta = \frac{1}{k_B T}$$

$$\lim_{\Delta \to 0} \cosh(\beta\Delta) = 1$$

$$\chi = -\frac{g_v g_s e^2 v^2}{6\pi c^2} \frac{1}{k_B T\left[\cosh\left(\frac{\mu}{k_B T}\right) + 1\right]}$$

susceptibility is $\sim \frac{1}{2k_B T}$ for $\mu = 0$ $(\mu \ll k_B T)$

but $\frac{1}{k_B T} e^{-\mu/k_B T} \to 0$ for $(k_B T \ll \mu)$

Essentially $\chi \approx 0$ for $\mu \neq 0$ and at very low temperature

[This is McClure's (Phys. Rev. 104 666 (1956)) result]

For $\mu = 0$ (neutral ideal graphene)

$$\chi = -\frac{g_v g_s e^2 v^2}{6\pi c^2} \frac{1}{2k_B T}$$ Curie-like diamagnetic susceptibility!

Standard Fermi-gas result (Landau diamagnetism, p. 70)

$$\chi = -\frac{1}{3} 2\mu_B \frac{m}{2\pi\hbar^2}$$ is T-independent!